Suggested teams and final projects

Computational and Bayesian Inverse Problems

PhD course, Linköping University, 26–30 October 2026

A companion to the participant booklet, compiled by Jan Glaubitz

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This document is for the participants of the course only. The suggestions are not assignments: you decide on your team and topic.

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About this document

This document suggests possible teams and final projects, built from the 19 participant profiles in the participant booklet. Page numbers in gray, such as “Andrae (p. 8)”, refer to the entries in the booklet.

What the suggestions are. They are starting points for the team formation on Tuesday, not assignments. Every suggestion starts from what you wrote in your profile: who you want to work with, whether you prefer a team or to work alone, which role you would like to play, and your own project idea. Where you stated a wish, the suggestion follows it; where you left the choice open, we propose one option and list alternatives on p. 11. If you prefer something else, that is fine.

Teams. We encourage teams of 2–4, but there is no hard restriction: working alone, or on a project that is only loosely related to the course, is fine too. Mixed teams of “problem owners” (model and data) and “method people” work well. Several teams below are built that way.

How each suggestion is described. Each block names a working title, the question the project could answer, the chapters of the lecture notes it builds on, a *minimum version* that is enough for a good report, a *stretch goal* for those who want to take the project towards a paper, the data, and the main risk. A good minimum version has a clear Bayesian formulation (prior, likelihood, posterior), at least one method from the course, and a validation on synthetic data with a known truth.

Default projects. Three default projects (A, B, C) come with a description and starter code. They are open to everyone, and some suggestions below build on them.

- **A:** variational Bayes for sparsity-promoting hierarchical models: hyper-priors, nonlinear forward models, and Bayesian neural networks.
- **B:** Bayesian atom-conserving chemical reaction neural networks for mechanism discovery.
- **C:** edge-preserving hierarchical priors for Bayesian bathymetry inference from shallow-water surface data.

Key dates.

- Tue 27 Oct, 13:15: thematic clusters and seed topics; team formation.
- Wed 28 Oct, 17:00: teams, working title, and one-paragraph proposal due (email to Jan).
- Thu 29 Oct, afternoon: 10-minute feedback meeting per team.
- Fri 30 Oct, 09:00: team pitches (3 minutes plus 2 minutes of questions).
- Tue 1 Dec, 23:59 CET: final report (8–15 pages, PDF) and link to the code repository.
- 7–11 Dec: 15-minute online report discussion per team.

Overview

	Members	Working title	Default
Teams			
T1	Andrae, Kalaivanan, Wikingsson	Removing the guidance bias of DAISI with interacting particles	–
T2	Ersing, Tymoshchuk (+ Habscheid)	Bayesian friction and sediment-transport parameters in Saint-Venant–Exner models	C
T3	Niemelä, Sefo	Bayesian speed-of-sound inversion: reduced-order versus full-order forward models	–
T4	Groppfeldt, Harushakamwe	Which defects are in the sample? Bayesian spectral unmixing	(A)
T5	Ghosh, von Klitzing	Bayesian atom-conserving reaction networks, or Bayesian calibration of a CVD model	B
Individual projects			
S1	Amadio	Surfactants on liquid-infused surfaces from pressure drops at several flow rates	–
S2	Dymér	Bayesian inference of frictional parameters from seismograms	–
S3	Haase	Uncertain initial conditions in the Euler equations: forward UQ and inference	–
S4	Hosseini Panah	Bayesian source strengths in airborne transfer path analysis	–
S5	Pilichos	Parameters of rotating Bose–Einstein condensates from density images	–
S6	Posysoev	Bayesian inverse Monte Carlo for effective interaction potentials	–
S7	Raab	Diffusion posterior sampling for black-hole imaging, checked against exact posteriors	–

Alternatives for the open choices (Habscheid, the Uppsala group, von Klitzing, Groppfeldt, and Harushakamwe) are on p. 11.

Suggested teams

T1: Removing the guidance bias of DAISI with interacting particles

Members: Martin Andrae (p. 8), Adhithyan Kalaivanan (p. 17), Erik Wikingsson (p. 25); the team you planned.
Default project: none needed
Course chapters: 3 (posteriors), 4 (MCMC and SMC, diagnostics), 5 (ensemble Kalman methods)

Question. DAISI conditions a learned generative prior on observations by guidance, which leaves an irreducible bias. Does interacting particle guidance remove it, and is the resulting posterior calibrated?

Minimum version. Measure the bias before you reduce it: start with a linear-Gaussian state-space model, where the Kalman filter and smoother give the exact posterior, and compare DAISI and DAISI with interacting particle guidance with it (means, covariances, CRPS, rank histograms).

Stretch goal. A low-dimensional chaotic system (e.g., Lorenz-96) with a long particle-filter or MCMC reference, then the weather-scale setting of your papers.

Data and risk. Synthetic data for the benchmarks. The main risk is scope: a small benchmark with an exact or reliable reference is worth more for the report than a large one without.

T2: Bayesian friction and sediment-transport parameters in Saint-Venant–Exner models

Members: Patrick Ersing (p. 10) and Kateryna Tymoshchuk (p. 24); optionally Jan Habscheid (p. 14) (see also p. 11).
Default project: C fits: your DG solver can replace the provided shallow-water model
Course chapters: 1 (sensitivity and identifiability), 3 (priors and MAP), 4 (MCMC), 5 (Laplace and ensemble Kalman methods)

Question. Which of the friction coefficient and the (about six) parameters of the sediment-discharge formula can be inferred from sparse, noisy observations of the free surface and the bed, and with what uncertainty?

Roles. Patrick and Kateryna bring the model and the structure-preserving DG solvers; Kateryna contributes the forward UQ and sensitivity analysis that she has in mind. Jan Habscheid would add sampling and hierarchical priors, which neither of you has used yet.

Minimum version. A one-dimensional bed-evolution test case with synthetic data and two or three parameters: a sensitivity and Jacobian analysis to see which parameters the data determine, then the MAP estimate, a Laplace approximation, and an MCMC reference.

Stretch goal. Infer the bathymetry jointly with the parameters, with an edge-preserving hierarchical prior (as in default project C), or use gradients through the solver for HMC.

Data and risk. Synthetic data and laboratory benchmarks. The main risk is the cost of the forward model inside a sampler: keep the grids coarse and the number of parameters small, or use an ensemble Kalman method or a surrogate.

T3: Bayesian speed-of-sound inversion: reduced-order versus full-order forward models

Members: David Niemelä (p. 19) and Jeremie Sefo (p. 23).
Default project: none needed
Course chapters: 3 (likelihood and model error), 4 (pCN, HMC), 5 (Laplace approximation)

Question. How does a posterior for the speed of sound that uses Jeremie's reduced-order model (ROM) as forward model differ from one that uses a full-order summation-by-parts (SBP) discretization, and can the Gauss–Radau error bounds of the ROM serve as a model-error term in the likelihood?

Roles. Jeremie brings the problem and the ROM; David contributes the SBP full-order solver as reference, as he asked to work on methods with other participants from Uppsala.

Minimum version. A one-dimensional acoustic problem with a low-dimensional or Gaussian prior on the speed of sound, synthetic data from a fine SBP solution, and the posteriors with the ROM and with the full-order model, both sampled with the same MCMC method.

Stretch goal. Treat the ROM error as an extra noise term, calibrated by the Gauss–Radau bounds, and check whether this restores the full-order posterior; two space dimensions.

Data and risk. Synthetic data. The two codes are in different languages, so agree early on a common interface (MATLAB is shared).

T4: Which defects are in the sample? Bayesian spectral unmixing

Members: Oscar Groppfeldt (p. 12) and Stephane Harushakamwe (p. 15).
Default project: none needed; the sparsity-promoting priors of default project A and Chapter 3 apply directly
Course chapters: 2 (nonnegativity and sparsity), 3 (hierarchical priors, IAS), 4 (MCMC), 5 (variational Bayes)

Question. An experimental absorption spectrum is modeled as a nonnegative combination of computed defect spectra plus model error and noise. Which defects are present, at which concentrations, and how certain are we, when formation energies rank the defects a priori?

Roles. Oscar brings the problem and the computed spectra; Stephane brings experience in regularization and estimation and would like to learn more estimation methods. Both of your research problems split an observation into factor contributions, so the methods transfer.

Minimum version. Synthetic spectra from computed defect spectra with known concentrations: nonnegative least squares and an ℓ^1 penalty as deterministic baselines, a Bayesian model with a sparsity-promoting hierarchical prior whose scale depends on the formation energy, and the posterior probability that each defect is present.

Stretch goal. Model the error of the computed spectra (shifts and broadening) as uncertain parameters; apply the same model to the attribution of value changes to market factors as a second application.

Data and risk. The experimental data cannot be shared, so the repository uses synthetic spectra. Overlapping defect spectra make some concentrations hard to separate; the posterior correlations then show which combinations the data determine, which is a result in itself.

T5: Bayesian atom-conserving reaction networks, or Bayesian calibration of a CVD model

Members: Aditya Prakash Ghosh (p. 11) and Christopher von Klitzing (p. 18).
Default project: B, or Aditya's CVD problem
Course chapters: 3 (priors), 4 (MCMC), 5 (Laplace approximation and variational inference)

Question. Option 1 (default project B): how much information does atom conservation add to the inference of a reaction mechanism, and can a Laplace approximation predict it? Option 2: which inputs of Aditya's CVD model (wall temperature, inlet turbulence, kinetic parameters) can be calibrated from deposition profiles, and with what uncertainty?

Roles. Aditya wants to contribute methods in a team and works on CVD precursor chemistry, close to the reaction kinetics of project B; Christopher prefers a small team, can take either role, and brings variational sampling. Project B comes with starter code, so you can start right away.

Minimum version. Option 1: the minimum tier of project B (verification on simplified cases, Laplace approximation against MCMC). Option 2: a polynomial chaos or Gaussian-process surrogate of the deposition profile from a moderate number of CFD runs, then MCMC on the surrogate with synthetic data, which turns the planned forward UQ into an inverse problem.

Stretch goal. Option 1: the paper track of project B, e.g., variational inference or normalizing flows for the posterior over mechanisms. Option 2: account for the surrogate error in the likelihood.

Data and risk. Synthetic data in both options. In option 2, the main risk is the cost of the CFD runs: fix the design of the runs early and keep the number of calibrated inputs to two or three.

Individual projects

These participants asked to work alone on a topic from their own research, or left the choice open and have a concrete topic. The suggestions sharpen your own idea; joining a team is of course possible too.

S1: Surfactants on liquid-infused surfaces from pressure drops at several flow rates

Members: Riccardo Amadio (p. 7).
Default project: none needed
Course chapters: 1 (SVD and information content), 3 (MAP), 4 (MCMC), 5 (Laplace approximation)

Question. With a model of the surfactant distribution that depends on flow-rate-independent parameters and on the flow rate Q, how much do pressure drops at several flow rates reduce the uncertainty in the surfactant amount, packing, and distribution?

Minimum version. An emulator (Gaussian process or polynomial chaos) of ΔP over the parameters and Q from a modest design of CFD runs, the MAP estimate and the Laplace approximation against MCMC on the emulator, and the posterior for one, two, and several flow rates.

Stretch goal. An explicit model-error term (e.g., a discrepancy function) and the emulator error in the likelihood; the meniscus shape as additional data.

Data and risk. Your simulations and measurements, shareable. The main risk is the number of CFD runs the emulator needs; fix the design early.

S2: Bayesian inference of frictional parameters from seismograms

Members: Robin Dymér (p. 9).
Default project: none needed
Course chapters: 3 (priors and noise model), 4 (pCN, MALA, HMC), 5 (low-rank Laplace approximation)

Question. Which combinations of the direct-effect parameter $a(x)$ do synthetic seismograms determine, and how do a derivative-free sampler and a gradient-based one with the adjoint gradient compare at equal cost?

Minimum version. A Gaussian or smoothness prior on $a(x)$, a Gaussian noise model, synthetic data, and the preconditioned Crank–Nicolson (pCN) sampler against MALA or HMC with the adjoint gradient, compared by effective samples per forward and adjoint solve.

Stretch goal. A low-rank Laplace approximation from your Hessian-vector products, used as a proposal or as a reference for the sampler; the prior-to-posterior variance reduction shows which directions the data inform.

Data and risk. Synthetic seismograms. The main risk is the cost of the forward solves: start with a small, coarse problem.

S3: Uncertain initial conditions in the Euler equations: forward UQ and inference

Members: Henry Morten Haase (p. 13).
Default project: none needed (A or C possible)
Course chapters: 3 (Bayesian formulation), 4 (MCMC), 5 (surrogates and approximate inference); Appendix A

Question. How do uncertain perturbations of the initial data (e.g., amplitude and wavelength of a Kelvin–Helmholtz instability) propagate through an Euler simulation, and can two or three of these parameters be inferred from a few noisy sensors?

Minimum version. Forward UQ with Monte Carlo and stochastic collocation in Trixi.jl, then a Bayesian inference of one or two initial-condition parameters from synthetic sensor data, with a surrogate built from the collocation runs.

Stretch goal. More parameters, structure-preserving (FSBP) discretizations as forward model, and how the posterior changes with the observation time.

Data and risk. Synthetic data, shareable. In sensitive regimes the posterior can be multimodal; short observation windows keep it tractable.

S4: Bayesian source strengths in airborne transfer path analysis

Members: Amir Hosseini Panah (p. 16).
Default project: none needed
Course chapters: 2 (Tikhonov, truncated SVD), 3 (Gaussian posterior), 4 (MCMC)

Question. In $\mathbf{p} = \mathbf{H}\mathbf{q} + \mathbf{e}$, with measured and therefore uncertain transfer functions $\mathbf{H}$, how much sound does each part of an electric drive unit generate, and how reliable are these estimates compared with matrix inversion?

Minimum version. Synthetic data with known sources: Tikhonov and truncated SVD as the conventional baselines, the Gaussian posterior with known $\mathbf{H}$, and how the credible intervals compare with the actual errors.

Stretch goal. Uncertainty in $\mathbf{H}$ as in the framework of Meggitt et al., which makes the problem nonlinear and calls for sampling; application to the measured data.

Data and risk. Volvo Cars owns the measured data, so the code repository should use synthetic data, and results on measured data go into the report only if Volvo agrees.

S5: Parameters of rotating Bose–Einstein condensates from density images

Members: Christos Pilichos (p. 20); alternatively in a team with David Niemelä or with the Uppsala group (p. 11).
Default project: none needed
Course chapters: 3 (Bayesian formulation), 4 (MCMC), 5 (surrogates, Laplace approximation)

Question. What do noisy density and vortex images tell us about the interaction strength and the rotation frequency, in particular near transitions between competing vortex configurations?

Minimum version. Two parameters, synthetic images from your finite element ground-state solver at low resolution, and the posterior on a grid of parameter values: with two unknowns, it can be computed and plotted directly, which shows multimodality without any sampler.

Stretch goal. A surrogate of the ground state in the parameters, MCMC with more parameters (e.g., the trap), and the uncertainty of derived quantities such as the number of vortices.

Data and risk. Synthetic data, shareable. Each forward evaluation is an energy minimization, so the number of evaluations is the bottleneck; near transitions the minimization may also land in different local minima.

S6: Bayesian inverse Monte Carlo for effective interaction potentials

Members: Maksim Posysoev (p. 21).
Default project: none needed
Course chapters: 2 (regularization), 3 (priors and MAP), 5 (Gauss–Newton and Laplace approximation)

Question. How uncertain is an effective interaction potential recovered from a radial distribution function $g(r)$, and does a prior stabilize the inverse Monte Carlo (IMC) iteration?

Minimum version. Use that the IMC cross-correlation matrix is the Jacobian of $g(r)$ with respect to the potential: with a Gaussian noise model and a smoothness prior, the IMC update becomes a regularized Gauss–Newton step, and its Hessian at convergence gives a Laplace approximation of the posterior, essentially for free. Test it on a reference simulation with a known potential.

Stretch goal. Propagate the posterior to other state points (temperature, density) to quantify transferability; compare with sampling in a low-dimensional parametrization.

Data and risk. Reference simulations. The statistical error of the simulated $g(r)$ enters the likelihood; estimate it from block averages.

S7: Diffusion posterior sampling for black-hole imaging, checked against exact posteriors

Members: Erik Raab (p. 22).
Default project: none needed
Course chapters: 2 (Tikhonov and TV), 3 (Gaussian posterior), 4 (MCMC and diagnostics)

Question. How well does diffusion posterior sampling recover black-hole images from sparse Fourier data, and how calibrated is its uncertainty?

Minimum version. Small synthetic images (e.g., ring-shaped sources) and a sparse Fourier sampling pattern, classical baselines (Gaussian posterior in closed form, Tikhonov and TV estimates), and a check of the sampler's bias against an exact posterior, e.g., with a Gaussian or Gaussian-mixture prior, for which the posterior is known.

Stretch goal. A diffusion prior trained on a larger set of simulated images, and coverage of credible intervals over many synthetic data sets.

Data and risk. Synthetic data. Training a diffusion prior takes time: use small images or a pretrained model.

Alternatives

Some of you left the choice of team or topic open. These alternatives are as good as the main suggestions; talk to each other on Tuesday.

Jan Habscheid.

- **With T2** (Ersing, Tymoshchuk): edge-preserving hierarchical priors for the bathymetry and the sampling, joined with their models and solvers; your coastline idea.
- **With Christopher von Klitzing on default project A**: variational Bayes for sparsity-promoting hierarchical models and for hierarchical Bayesian neural networks, which joins your BNN research with Christopher's variational sampling. Aditya Ghosh would then take project B or his CVD calibration (T5), alone or with another participant.

The Uppsala group (Niemelä, Pilichos, Sefo).

- **Main suggestion:** Niemelä and Sefo as T3; Pilichos individually (S5).
- **Niemelä and Pilichos:** inference for rotating condensates (S5) with David contributing the numerics; Sefo individually on his ROM-based stochastic inversion.
- **All three:** Sefo's speed-of-sound problem (T3), with Pilichos contributing the sampling and the parameter study.

Christopher von Klitzing. Instead of T5 or project A, an own topic: uncertainty quantification for an implicit transfer operator model on a two-dimensional toy potential (e.g., with an ensemble of models or a Laplace approximation), propagated to the predicted transition rates.

Oscar Groppfeldt and Stephane Harushakamwe. If you prefer your own topics, both work alone: Oscar with the spectral unmixing of T4; Stephane with a Bayesian version of Blomvall's convex model for interest-rate measurement, where the convex optimization is the MAP estimate under a smoothness prior and the posterior gives the uncertainty of the forward rates and of the attribution built on them.

Anyone. The default projects A, B, and C are open to everyone, as a team or individually.